



DEBARK UNIVERSITY
COLLEGE OF BUSINESS & ECONOMICS
DEPARTMENT OF MANAGEMENT

Business Research Methods Module for Exit Exam

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Course Objectives

- ❖ Understand the role of business research
- ❖ Differentiate different types of research
- ❖ Identify and define research problem
- ❖ Design appropriate research design know sampling design
- ❖ Differentiate types of data gathering methods, Analyze and interpret data
- ❖ Prepare standard research project



Chapter One - Research Methods: An Introduction

Learning objectives

- Describe meaning and purpose of research.
- Differentiate between different types of research.
- Understand about Motivation for Research
- Describe criteria for good Research

Meaning of research

In fact, research is the art of scientific investigation. The advanced learner's dictionary lays down the meaning of research as a careful investigation or inquiry especially through search for new facts in any branch of knowledge.

The dictionary defines the *re* as a prefix meaning ***again***, or ***over again*** and the ***search*** as a verb meaning to *examine closely and carefully, to test or to probe*. Together they form a noun, ***research***, which means careful, systematic, patient study and investigation in some field of knowledge, undertaken to establish facts or principles.

Redman and Mory define research as a systematic effort to gain new knowledge. Some people consider research as a movement from the known to the unknown. It is actually a voyage of discovery.

D.Slesinger and M. Stephenson define research as the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or other purpose. Research is thus an original contribution to the existing stock of knowledge making for its advancement.

The word ***research*** is composed of two words- *re* and *search*.

Research is a systematic process of collecting and analyzing data in order to increase our understanding of the phenomenon with which we are concerned or interested.

Research is the systematic, objective, empirical and controlled process of gathering, recording and analyzing data for aid in making scientific and critical decision in discovery of new knowledge.

Research refers to the systematic method consisting of stating the problem, formulating hypotheses, collecting the facts or data, analyzing the facts and reaching on conclusions that is in the form of solutions towards the concerned problem or in certain generalizations for some theoretical formulation.

Objectives of Research

1. To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as exploratory or formulative research studies);
2. To portray accurately the characteristics of a particular individual, situation or a group (studies with this object in view are known as descriptive research studies);
3. To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies);
4. To test a hypothesis of a causal relationship between variables (such studies are known as hypothesis-testing research studies).

Motivation in Research

The possible motives for doing research may be either one or more of the following:

- Desire to get a research degree along with its consequential benefits;
- Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiates research;
- Desire to get intellectual joy of doing some creative work;
- Desire to be of service to society;
- Desire to get respectability

Types of research

There are different ways of classifying research in terms of:

1. Based on the Goal: basic and applied research
2. Based on design: descriptive, explanatory and exploratory
3. Based on Approaches: quantitative, qualitative & mixed
4. Based on way the research is conducted: experimental and non-experimental
5. Based on time- cross-sectional and longitudinal

Classification of Research based on the Goal of Research

The goal of research is problem solving. The nature of the problem that the research attempts to solve could be theoretical or practical – building a theory or solving immediate practical problems. These two types of problems that the research tries to solve leads to two broad classifications of research:

- Basic (fundamental or pure) research and Applied research

I. Pure/Basic Research

Basic research its primary objective the advancement of knowledge and the theoretical understanding of the relations among variables. It is basically concerned with the formulation of a theory or a contribution to the existing body of knowledge. That is, basic research is designed to add to an organized body of scientific knowledge and does not necessarily produce results of immediate practical value.

The major **aims** of basic research include:

- **Discovery:** where a totally new idea or explanation emerges from empirical research which may revolutionize thinking on that particular topic.
- **Invention:** where a new technique or method is created.
- **Reflection:** where an existing theory, technique or group of ideas is re-examined possibly in a different organizational or social context.

The motivation behind is to **expand human knowledge**, not to create or invent something that has practical significance.

II. Applied Scientific Research - Applied research is designed to solve practical problems of the modern world, rather than to acquire knowledge for knowledge's sake. It is undertaken to solve immediate practical problem and the goal of adding to the scientific knowledge is secondary.

Classification of Research based on Designs

Once the researcher has determined the specific question to be answered and has operationalized the variables and research question into a clear, measurable hypothesis, the next task is to consider a suitable research design. They usually fall into the following three general categories: Exploratory, Descriptive and Explanatory research

Descriptive research: its goal is description of the existing phenomena. It presents a picture of details of a situation, social setting or relationships. It answers the question of ‘what is happening now?’ “E.g. what is the status of car accident in Godar? It looks characteristics of individuals, groups, institutions in present state area of investigation to obtain precise information in varies field of inquiry. It helps explain the study based on researchers knowledge.

Exploratory research: is used when little is known or no scientific knowledge is available about groups, activities, processes and situations. It is the initial research. The goal may be to formulate questions that future research may answer. It is usually employed in qualitative research like case study, grounded theory etc. It is used in pilot study research

Explanatory research: involves explanation, or determining why or how something occurred. It would go beyond descriptions, to analyze and explain *why or how* something is happening. Thus, explanatory or analytical research aims to understand phenomena by discovering and measuring causal relations among them. That is, explanatory research looks for causes and reasons. e.g the effect of human resource management practice on operational performance in case of Dashen Brewery Factory.

Classification based on Approaches of Research

Qualitative research is an approach for exploring and understanding the meaning individuals or groups ascribe to a social/human problem. The process of research involves emerging questions and procedures, data typically collected in the participant's setting, data analysis inductively building from particulars to general themes, and the researcher making interpretations of the meaning of the data. The final written report has a flexible structure. Those who engage in this form of inquiry support a way of looking at research that honors an inductive style, a focus on individual meaning, and the importance of rendering the complexity of a situation.

Quantitative research is an approach for testing objective theories by examining the relationship among variables. These variables, in turn, can be measured, typically on instruments, so that numbered data can be analyzed using statistical procedures. The final written report has a set structure consisting of introduction, literature and theory, methods, results, and discussion. Like qualitative researchers, those who engage in this form of inquiry have assumptions about testing theories deductively, building in protections against bias, controlling for alternative explanations, and being able to generalize and replicate the findings.

Mixed methods research is an approach to inquiry involving collecting both quantitative and qualitative data, integrating the two forms of data, and using distinct designs that may involve philosophical assumptions and theoretical frameworks. The core assumption of this form of inquiry is that the combination of qualitative and quantitative approaches provides a more complete understanding of a research problem than either approach alone.

Based on the way the research is conducted

A) **Experimental research:** it is a research carried out by designing and conducting an experiment. The different kinds of experimental research are:

- i) True experimental research- In this kind of research participants are assigned randomly to experimental treatments and research should also be replicable.

- ii) **Quasi-Experimental research:** this is similar to true experimental research. However, instead of having participants randomly assigned to experimental treatments, only naturally assembled groups are used in the research.

B) Non- Experimental research

It differs from the experimental research one in the amount of manipulation of subjects. It also differs in the level to which the investigator controls that he/she studies.

The different kinds of Non-experimental research are:

- i) **Survey research-** This involves obtaining a general picture about the number of people who have the knowledge or experience with the phenomenon or the problem in question. It is the most widely used research. It involves setting questions and interviews. There are many approaches to interviews. The main ones are;
 - a) Personal interview
 - b) Telephone interview and
 - c) Mail interview
- ii) **Historical research:** it consists of studying a problem or an issue or a phenomenon or movement in the past. It deals with things that have already happened and the researcher searches for data rather than producing new. It is concerned with describing what was” rather than “what is”.
- iii) **Ethnographic research:** is an in depth analytical description of a specific cultural situation. It takes place in natural setting and focuses usually on the processes in attempt to obtain a holistic picture.

Classification based on time- cross sectional and longitudinal research.

A **cross-sectional study** examines a particular phenomenon or observations at a specific period of time and is mostly adopted by academic researchers due to time constraints.

Longitudinal studies in which multiple observations are made over long period of time or two or more samples are studied at different period of time (Kothari, 2014). It is used to find the pattern of change in relation to time. The study population is contacted on multiple occasions at regular intervals, typically over a lengthy period of time, to gather the necessary data.

Research process

- The research process involves the following steps.

1. Formulating research problem

2. Review of literature
 - Review of concepts and theories
 - Review of previous research findings
3. Formulating, hypothesis/ questions and objectives
4. Designing the research
5. Developing instrument for data collection
6. Collection of data
7. Analysis of data
8. Writing research report

Purposes of Research

In the business arena, research is required because of the following reasons:

- To identify and find solutions to the problems
- To help in making decisions
- To develop new concepts
- To find alternate business strategies
- To gain a competitive advantage.
- To test new products and services.
- To enhance profitability.
- To reduce operational costs.

Why Research is conducted?

- Research is conducted for a number of reasons, which in turn depend on the objectives of any particular '*research problem*'.
- In the business area, however, research tends to be undertaken in order to achieve one or more of the following objectives:
 - To gain a competitive advantage and to test new products and services.
- To solve a management/organizational problem.
- To provide information which may help to avoid future business problems.
- To forecast future sales.
- To better understand shifts in consumer attitudes and tastes.
- To enhance profitability and to reduce operational costs.
- To enable management to prioritize strategic options for the future.

Criteria of good research

Whatever may be the types of research works and studies, one thing that is important is that they all meet on the common ground of scientific method employed by them. One expects scientific research to satisfy the following criteria:

1. **Good research is systematic:** It means that research is structured with specified steps to be taken in a specified sequence in accordance with the well-defined set of rules. Systematic characteristic of the research does not rule out creative thinking but it certainly does reject the use of guessing and intuition in arriving at conclusions.
2. **Good research is logical:** This implies that research is guided by the rules of logical reasoning and the logical process of induction and deduction are of great value in carrying out research. Induction is the process of reasoning from a part to the whole whereas deduction is the process of reasoning from some premise to a conclusion which follows from that very premise. In fact, logical reasoning makes research more meaningful in the context of decision making.
3. **Good research is empirical:** It implies that research is related basically to one or more aspects of a real situation and deals with concrete data that provides a basis for external validity to research results.
4. **Good research is replicable:** This characteristic allows research results to be verified by replicating the study and thereby building a sound basis for decisions.

Chapter Two

Defining Research Problem and Hypothesis Formulation

Chapter Objectives

After completing this unit, students will be able to:

- ✚ Define research problem
- ✚ Identify Components of research problem
- ✚ Formulate research Problem, hypothesis and research question
- ✚ Identify Criteria for hypothesis formulation

What is a Research Problem?

In research process, the first step happens to be that of selecting and properly defining a research problem. Problem means a question or an issue to be examined or description of an issue currently existing which needs to be addressed. A research problem refers to some kind of problem which a researcher experiences or observes in the context of either a theoretical or practical situation. The researcher has to find out suitable course of action by which the objective can be attained optimally in the context of given environment. Thus, selection of research problem has high value to the society and the researcher must be able to identify those problems that need an urgent solution.

It provides the context for the research study and generates the questions which the research aims to answer. The statement of the problem is the focal point of any research. The statement of the problem involves the demarcation and formulation of the problem, i.e., the who/what, where, when, why.

Research problem formulation requires knowledge of a given field of study and research methodology.

- Example of research problems
 - *Why public manufacturing companies in Ethiopia are unprofitable?*
 - *Why prevalence of corruption is high in SSA Countries?*
 - *What are the challenges of federalism in Ethiopia?*
 - *What are the investment opportunities of Debarq city?*

A **problem** occurs when there is a difference between the current conditions and a more preferable set of conditions. The gap can come about in a number of ways:

1. Business performance is worse than expected business performance. For instance, sales, profits, and margins could be below targets set by management. This is a very typical type of problem analysis. Think of all the new products that fail to meet their targeted goals. Management is constantly monitoring key performance variables. Previous performance usually provides a benchmark forming expectations. Sales, for example, are generally expected to increase a certain percentage each year. When sales fall below this expectation, or particularly when they fall below the previous year's sales, management usually recognizes that they have a potential problem on their hands.
2. Actual business performance is less than possible business performance. Realization of this gap first requires that management have some idea of what is possible. This may form a research problem in and of itself. Opportunity-seeking often falls into this type of problem-definition process. Many American and European Union companies have redefined what possible sales levels are based upon the expansion of free markets around the world. China's Civil Aviation Administration has relaxed requirements opening the Chinese air travel market to private airlines. Suddenly, the possible market size for air travel has increased significantly, creating opportunities for growth.

Examples of Well- defined problems:

- "The frequency of job layoffs is creating fear, anxiety, and a loss of productivity in middle management workers."
- What factors were responsible for the lower labour productivity of Ethiopia's manufacturing industries during the time 20-to 20-relative to other African countries?
- To what extent has the new advertising campaign been successful in creating the high quality, customer-cantered corporate image that it was intended to produce?
- What are the effects of downsizing on the long-range patterns of companies?
- Does expansion of internal operations result in an enhancement of the firm's image and value?

Components of Research Problem

There are a number of elements (components) which a problem must have before it becomes a research problem ready for study.

1. Objective or aim of the problem which is to be investigated. This answers the question "Why?" Why is there a need for investigation, inquiry or study?

2. The topic or theme which needs to be investigated. This answers the question “What?” What is to be researched or studied?” For example: What would a rival company do if we decrease our prices by 25%? What would sales be if prices were \$ 89? \$. 99? How would a rival firm’s action influence our sales and profits?

3. The time dimension of a decision problem is always the future: The period or time of the study when the data are to be gathered. This answers the question “When?” When is the research to be performed?” Managers frequently run the risk of making the correct decision at incorrect time. It is essential that the decision maker as well as the researcher determine the right time reference for-the decision.

4. The area or location in which the study is to be conducted. This answers the question “Where?” Where we need to conduct the study? The space coordinates give you the geographic boundaries within which the action is to be taken. In the problem definition, these lines are hardly ever neat political divisions or subdivisions. The universe of interest should be defined either conceptually or by enumeration.

5. Population or universe from whom the data needs to be gathered. This answers the question “Who?” or “from whom?” Who are the respondents? From who are the data to be collected?” They may include persons, groups of persons, business establishments.

Problem definition process

The technique of defining a research problem involves the following steps.

(i) Statement of the problem in a general way: which must be resolved by cool thinking and rethinking over the problem.

(ii) Understanding the nature of the problem: to understand **its origin and nature clearly.** The researcher should also keep in view the environment within which the problem is to be studied and understood.

(iii) Surveying the available literature: “Knowing what data are available often serves **to narrow the problem itself** as well as the technique that might be used.

(iv). developing the ideas through discussions: Discussion concerning a problem often produces useful information. Hence, a researcher must discuss his problem with his colleagues and others who have enough experience in the same area on similar problems. They help him sharpen his **focus of attention on specific aspects** within the field.

(v) Rephrasing the research problem: Finally, the researcher must sit to rephrase the research problem **into a working proposition**. the researcher puts the research problem in as specific terms as possible so that it may become operationally viable and may help in the development of working hypotheses.

Choosing Variables to Study

Researchers attempt to understand how and why differences in one variable are related to differences in another variable. research study is identifying what variables will be the focus of the study

Independent Variables vs. Dependent Variables

- the independent variable is what causes or influences the outcome
- The dependent variable is called “dependent” because it is influenced by the independent variable
- **Extraneous variables** are independent variables that are not related to the purpose of the study, but may affect the dependent variable.
 - *Example: if the researcher wants to study the effect of commitment on employee performance, motivation, co-workers' support, work environment, can be other variables that affect employee performance.*
- There is no restriction on the number of variables that can be included in a problem statement; however, model parsimony should be maintained.

Categorical Variables vs. Continuous Variables

- **Categorical variables** are variables that can take on specific values only within a defined range of values e.g. gender
- **continuous variables** are variables that can theoretically take on any value along a continuum e.g. age

Research questions

Stating research question is the starting point of the point in problem identification. A research question is the researcher’s translation of the business problem into a specific need for inquiry.

A clear and concisely stated research question is the most important requirement for a successful study. A research question is presented in question format and have the following benefits

- Detail the problem statement
- Further describe and refine the issue under study

- Add focus to the problem statement
- Guide data collection and analysis

Example 1: “The problem is to determine the best ways our company can train existing and potential users of networked personal computers.” (Problem Statement)

This problem statement led to the following research questions:

RQ1: How familiar are employees with the various software applications?

RQ2: what attitudes do employees have toward these software packages?

RQ3: How important are the various factors for evaluating the use of a PC?

RQ4: How effective are training efforts in increasing knowledge and use of the new applications?

Research Hypothesis

Hypothesis is defined as a set of proposition set forth as an explanation for the occurrence of some specified group of phenomena either asserted merely as a provisional conjecture to guide some investigation or accepted as highly probable in light of established facts. A research hypothesis is a predictive statement, capable of being tested by scientific methods, that relates an independent variable to some dependent variable. Hypothesis should always be written before the study and should not be changed after the study results are examined.

Example:

1. Age of tax payers will not contributes to the tax evasion practice.
2. Study and CGPA of students are positively related.
3. Profitability and asset management have no relationship.
4. Project planning has a positive effect on project cost
5. Nutrition and metal health have direct relationship.

Characteristics of Good Hypothesis

- Hypothesis should be clear and precise
- Hypothesis should be capable of being tested
- Hypothesis should be limited in scope and must be specific
- Hypothesis should state relationship between variables
- Hypothesis should be consistent with most known facts
- Hypothesis must actually explain what it claims to explain

Statements of Hypotheses: Formats

- **If-then format**
 - If the students study hard, then they will pass the exam.
 - If moral standards are thought at childhood, then corruption will reduce in Ethiopia.
- **Directional Hypothesis**
 - Asset management has positive impact on firm performance.
 - Diet has direct relationship with mental health
- **Non directional hypothesis**
 - Asset management has relation with firm performance
 - Diet has relation with mental health.

Null Hypothesis and Alternative Hypothesis

- **Null Hypothesis** is a statement that there is no significant difference or relationship or association between variables.
 - Example: High income tax rate not contributes to tax evasion.
- **Alternative Hypothesis** is a statement that there is significant difference or relationship or association between variables
 - Example: High income tax rate contributes to tax evasion.

Importance of Hypothesis

In research, a hypothesis serves several important functions:

- It guides the direction of the study
- It identifies facts that are relevant and those that are not:
- It suggests which form of research design is likely to be most appropriate:
- It provides a frame work for organizing the conclusions that result.

Criteria for hypothesis formulation

It depends upon a researcher's knowledge abilities which lead to definition of hypothesis. there are certain suggestions enumerated below which could assist in developing hypothesis.

a. **Induction by Simple Enumeration:** We observe that all birds, irrespective of size, have beaks and wings. From this observation, a researcher can form the hypothesis that birds are creatures with wings and beaks.

b. **Method Agreement:** In this approach, various objects forming a group to have a common circumstantial background can lead to development of a hypothesis. For example, a doctor

discovers that all lung cancer patients are chain smokers, then he can easily formulate a postulate that chain smoking leads to throat cancer. Another example could be that, if passengers in a bus felt giddy then it could be concluded that toxic fumes entered the passengers' compartment. Therefore, hypothesis can be relevant to a problem if it expresses determination of connection between a set of facts.

c. **Analogy:** Sometimes certain commonalities between various things can lead to formulation of hypothesis. For example, if we find that many people from North Eastern States have slanted eyes, we can hypothesize that all other North Eastern people also have slanted eyes because they belong to North East. Such a hypothesis based on a phenomenon called analogical reasoning.

d. **Concomitant Variation:** At times, hypothesis can be formulated about two phenomena, by correlating relationship between them. For example, if we find that chocolate in attractive wrappers sells more than those in simple wrappers, a hypothesis can be formulated that chocolates if wrapped attractively will sell more.

e. **General Culture:** The general culture pattern facilitates in formulating a hypothesis, and also to guide its trend. The mythological basis in Indian culture may form a suitable basis for hypothesis of a religious research.

f. **Scientific Theory:** Many times, it is the scientific theory that provides underpinning of a hypothesis. Such scientific theories enable researchers to formulate hypothesis which could be extension or corollaries of such scientific theories.

g. **Personal Experience:** Sometimes the facts are there, but a right individual sees it in the right perspective and formulates a hypothesis.

Research Objectives

The research objective is the researcher's version of the business problem. Objectives explain the purpose of research in measurable terms and define standard of what the research should accomplish. Objectives should be simple, specific, stated in advance, stated using action verbs. Objectives can be classified into:

- General objective – showing what exactly to be studied, and states the desired outcomes from the study.
- Specific objectives – shows in greater detail the specific aims of the research project.

The objective(s) should be phrased in positive terms (e.g., to develop, to determine, to measure, to investigate, to study)

Example 1

“The problem is to determine the best ways our company can train existing and potential users of networked personal computers.” (Problem Statement stated as a broad/general objective)

RO1: To identify how far employees are familiar with the various software applications

RO 2: To describe employees’ attitudes towards the available software applications

RO 3: To determine the importance of factors that affect -----

RO 4: To measure the effectiveness of the various training methods/efforts in increasing the knowledge and use of new applications

Chapter Three

Research Proposal

Learning Objective: after completing study on this chapter, you should be able to;

- Define research proposal
- Identify purpose of Research Proposal
- Identify elements of Research Proposal
- Prepare research proposal

What is research proposal?

A research proposal is a plan of action for the actual research to be conducted. It is a selling document not the technical report. The objective in writing a proposal is to describe what you will do, why it should be done, how you will do it and what you expect will result. It is a document which sets out your ideas in an easily accessible way.

Research proposal can be defined as a written document requesting both authorization and funds to undertake a specific research project. It always includes a statement explaining the purpose of the study or a definition of the problem. It systematically outlines the particular research methodology and details the process that will be utilized at each stage of the research process.

Purposes of Research Proposal

- A research proposal is intended to convince others that you have a worthwhile research project, the competence and the work-plan to complete it.
- To demonstrate expertise in a particular area of study
- To demonstrate competency in a particular area of study
- To serve as a contract and planning tool

Importance of Research Proposal

The research proposal has the following importance.

- It allows the sponsor to **assess the value** of the research and **competency** of a researcher,
- It enables the sponsor to **evaluate the** proposed **design** and its **appropriateness**,
- It serves as a **framework for discussion** and further **refinement** between the sponsor and researcher.
- It serves as a basis for determining the feasibility of the research project.
- It enables the researcher **to plan** and **review** the **project steps**, it is a guiding manual
- Proposal forces **time** and **budget** estimates

A proposal tells us:

- What will be done
- Why it will be done
- How it will be done
- Where it will be done
- To whom it will be done, and
- What is the benefit of doing it?

Elements of Research Proposal

The following major components:

Preliminary parts

- *Title page*
- *Table of contents*
- *Acronyms*
- *Acknowledgements*
- *Abstract*

Main Body

I. Introduction

- 1.1 Background of the study*
- 1.2 Statement of the problem*
- 1.3 Research Questions/Hypothesis*
- 1.4 Objectives of the study*
- 1.5 Significances of the study*
- 1.6 Delimitations /scope of the study*
- 1.7 Definitions of terms*

II. Literature Review

III. Methodology of the study

- 3.1 variables of the study*
- 3.2 Research Strategies*
- 3.3 Sources of data*
- 3.4 Methods of sampling*
- 3.5 Methods of Data Collection*
- 3.6 Methods of Data processing and analysis*

IV. Work plan

V. Cost Budget

Ending Matters

- *References*
- *Appendix*

1. Title page

This is the first page of the proposal document and contains the title of the thesis, the author's name, and affiliation, name of the advisor and the date and place of submission.

Title of the thesis:

- ✓ The title should summarize the main topic of the study.
- ✓ It should clearly state the purpose of the study.
- ✓ The title should be written in a short and precise manner and should be fully explanatory.
- ✓ Do not use abbreviations in a title. Spell out all terms.
- ✓ **Author's name:** Indicate your full name including your ID number.
- ✓ **Affiliation:** Indicate the name of your school/institution and your status as a student.
- ✓ **Name of advisor:** Indicate the name of your thesis advisor.
- ✓ **Date and place:** At the end of the title page, state the date of your proposal and the name of the city where your institution is situated (Debarq, Gondar, Bahir Dar, Addis Ababa, ...Ethiopia).

2. Abstract

At the top of the page, centered, you should have the word "Abstract." It is a concise summary of the entire paper, including the problem, major hypotheses, objectives, sample and population, a brief description of the measures, and the name of the design or a short description. Is summary, which reflects the whole content of the proposal (most of the time less than 300 words?). Abstract should be concise, informative and should provide brief information about the whole problem to be investigated

3. Background of the study

- Under this section, discuss briefly the literature related to the study. However, note that you are not expected to include exhaustive historical review.
- Introduce the problem briefly.
- Summarize earlier related works in brief, avoiding non-essential details.
- Focus on brief and pertinent ideas, methods, findings and conclusions.
- State briefly the purpose of your study in a general format, connecting your brief review of the literature with the problem you intend to investigate.

4. Statement of the problem

- States the issue that leads to a need for the study.
- It answers the question 'why does this research be conducted?'

- The foundation for everything to follow in the proposal.
- A clear, concise statement of the problem to be solved by the proposed research, usually in few sentences.
- The problem provides the context for the research study and typically generates questions which the research hopes to answer.
- The problem statement should “hook” the reader and establish a persuasive context for what follows.
- The problem statement should close with research question or objectives.
- Discuss the background, efforts made so far, possible causes, and consequences of the problem.

5. Research Question

- The research questions can be embedded in the statement of the problem or treated under a separate section.
- Your research questions may focus on description of what the situation, process and/or magnitude of the problem is; explanation of why the problem occurred; and prediction of what will happen next if the problem continues or discontinues.
- A clear and concisely stated research question is the most important requirement for a successful study. *examples*
 - ✚ What are the causes of poor quality of education in Ethiopia?
 - ✚ What is the relationship between profitability and sources of finance?
 - ✚ Is tax evasion is related to the educational background of taxpayers?
 - ✚ How to promote saving habits of university students?
 - ✚ What are the constraints of SME in Addis Ababa city?
 - ✚ What is the attitude of people towards insurance services in Ethiopia?

6. Objectives of the study

Aim which enquirer seeks to bring about as a result of completing the research. As research questions are generated from the research topic or research idea/problem, research objectives can be driven from the research questions. Objectives can be classified into:

- General objective – showing what exactly to be studied, and states the desired outcomes from the study.
- Specific objectives – shows in greater detail the specific aims of the research project.

7. Hypothesis (Optional)

Hypotheses are usually generated from the research problem or questions. They should be specific and testable. Hypotheses are subject to acceptance or rejection following the result of the data.

Example

- **Research title:** The impact of procurement on production efficiency.
- **Research question:** To what extent procurement is related to production efficiency?

Research objective:

General objective - To determine the influence of procurement on production efficiency.

Specific objectives- To examine the degree of procurement influence on production efficiency.

- **Hypothesis-** Procurement has positive effect on efficiency of production.

8. Significance of the proposed study

- ❖ Indicate what will be the use of the research output and how it will be used.
- ❖ Specify whether your study will have **practical** or **theoretical** significance or both.
- ❖ If the study intends to empirically inform decision makers and others concerned about the problem at hand; to suggest ways and means of tackling the problem at hand; to fill in current gaps specified by the problem; to shed light for further studies; etc., it will have practical significance.
- ❖ In this section outline the relevance of the study from policy as well as academic view points

9. Delimitation/Scope of the Study

- In this section, you are expected to clearly describe the boundary of your research.
- The scope of your research can be delimited in terms of geographical location, variables, and subjects to be used. You should, therefore, clearly state the delimitation of your study and specify the reasons thereof.
- Remember that scope or delimitation is usually induced due to resource and skill constraints (money, time, experience, etc.).

10. Definition of terms

Relevant concepts or terms should be precisely defined in this section.

Conceptual definition of terms: refer to the definition of concepts/terms or variables from the theoretical perspective. Sources of such definitions should be cited.

Operational definition of terms: include practical definition of terms given by the researcher as they appear to be in the text. If you are sure that your definition of terms are operational, i.e., if they are your own definitions, there is no need of mentioning any source.

11. Literature review

A literature review is a description of the literature relevant to a particular field or topic. The literature review asks how similar and related questions have been answered before. It gives an overview of what has been said, who the key writers are, what are the prevailing theories and hypotheses, what questions are being asked, and what methods and methodologies are appropriate and useful.

Importance of Literature Review

- Gives credits to those who have laid the ground work for your research.
- Demonstrates your knowledge of the research problem.
- Demonstrates your understanding of the theoretical and research issues related to your research question.
- Shows your ability to critically evaluate relevant literature information.
- Indicates your ability to integrate and synthesize the existing literature.
- Provides new theoretical insights or develops a new model as the conceptual framework for your research.
- Convinces your reader that your proposed research will make a significant and substantial contribution to the literature (i.e., resolving an important theoretical issue or filling a major gap in the literature).

12. Research Methodology

The methods section is an outline of all the activities that the researcher is planning to perform in the pursuit of his/her research goals. The methods section includes a descriptions of the research design, study participants, measuring instruments, sources and types of data, procedures and data analysis methods.

Research design- Indicate what type of research you are undertaking and how it is designed. Research design usually refers to the blueprint of the research. It is similar to the design of a building developed by an architect in which the building is constructed accordingly.

Method of data collection: How the data are obtained from earlier specified sources? There are different methods to collecting primary data. All available methods are expected to be considered

by a researcher and the best one is selected. The researcher explains and justifies how his/her selection of a particular method is relevant and appropriate to his/her study.

Method of sampling: From which elements will the data be collected? The researcher has to specify the elements from which the desired information is collected and how these elements are selected. There are different sampling techniques. The researcher is expected to consider all of them and select the one which is appropriate for the current study. Justification of his/her choice of a particular sampling technique is relevant here. The sample size should also be specified with its method of sample size determination.

Method of analysis: how data will be processed and analyzed should be explained in this part. Particularly if the researcher uses mathematical models like simple or multiple regression, time series or liner programming models, he/she has to show their mathematical derivations and relations briefly to demonstrate their relevancy for the current study.

Method of presenting the outcome: Sometimes it is worth to mention how the outcome of the research will be presented to consumers (general public). This is important if the research is not a part of a particular study program (like, master or doctoral study).

13. Work Plan and Budget

1. Work plan

A timetable explaining how the researcher expects to carry out his/her project and when each of the important phases will be completed is helpful to both the researcher and the reviewer. It is a plan in terms of numbers of weeks or months and expected completion dates. Commonly researcher presents it in a table form.

The length of time required for postgraduate study (Doctoral study) is determined by the budget of the time presented by students. Thus, researcher should be realist in budgeting his time.

Work Plan

No.	Activities	Time in weeks											
		December				January				February			
		1	2	3	4	1	2	3	4	1	2	3	4
1	Preliminary review												
2	Preparation of proposal												
3	Submitting proposal												

4	Questionnaire and interview design		
5	Data collection		
6	data editing and verifying		
7	Presentation and analysis		
8	Finalizing first draft		
9	Revision and editing		
10	Printing final draft		
11	Submission and presentation of report		

2. Cost budget

Most proposals put together with the expectation that funding will be necessary and an itemized list of items needed to carry out the methodology is listed in some detail. A carefully developed budget reflects the seriousness of the proposal and the degree to which it is a realistic assessment of what is needed.

Budget

No	Item	Unit	QNTY	Unit Cost	Total Cost	Remark
1	Pen	Number	5	2.00	10.00	
2	Pencil	Number	3	1.00	3.00	
3	Paper	Pad	4	80.00	320.00	
4	Computer	Number	1	3000.00	3000.00	Leasing
5	Transportation	-		300.00	300.00	Data Collection related
6	Printing	Page	300	1.00	300.00	
7	Photocopy	Page	100	0.50	50.00	Questionnaire related
8	Secretarial Service	Page	100	2.00	200.00	
9	Internet Service	Hours	30	8.00	240.00	
10	Miscellaneous			2000.00	2000.00	
Total					6423.00	
Contingency (10%)					642.30	
Grand Total					7065.30	

14. Reference (Bibliography)

Indicate the reference materials or sources you consulted in preparing your research proposal.

- **References:** are lists of literature, which have been cited in the text of a document
- **Bibliography:** includes also items, which were not cited but are relevant to the document.

- You must give references to all the information that you obtain from books, papers in journals, and other sources.
- References may be made in the main text using authors name (Harvard style). End of the sentence = (Tofik, 2015), At the beginning of the sentence= According to Tofik (2015)
- You will also need to place a list of references, numbered as in the main text (or alphabetically ordered), at the end of your research proposal.
- *American Psychological Association (APA) style guide. Author (Year of publication), Title of the source(italic / underline), Edition number, name of publisher, the page numbers for your reference*

Examples using

Harvard style

Kothari, C.R., 2004. *Research methodology: Methods and techniques*. New Age International.

Jharotia, A.K. and Singh, S., 2016. Use of Research Methodology in Research: An Overview. *International Journal of Social Science, Journalism & Mass Communication*, 2(2), pp.44-51.

APA

Kothari, C. R. (2004). *Research methodology: Methods and techniques*. New Age International.

Jharotia, A. K., & Singh, S. (2016). Use of Research Methodology in Research: An Overview. *International Journal of Social Science, Journalism & Mass Communication*, 2(2), 44-51.

Chapter Four

Research Design

Learning objectives: After studying this chapter you should be able to:

- Understand concepts of research design
- Identify features of Research Design
- Distinguish different types of research design

Meaning of Research Design

- Research design is the arrangement of condition for collection and analysis of data in a manner that aims to combine relevance to the research purpose. Research design is a procedural plan that is adopted by researchers to answer questions objectively, accurately, economically and with validity.
- Research design is a plan for selecting the **sources and types of information** used to answer research questions and/or to test the hypothesis. A framework for specifying the relationships among the study variables.
- A **blueprint/detail plan** that outlines each procedure from the formulation of hypothesis to the analysis of data. Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study.

Need for research design

- It facilitates the smooth operation of various research activities
- It makes research as efficient as possible yielding maximum information with minimum expenditure
- It helps to organize ideas
- Design can be given to others for their comments and critical evaluation. In the absence of such a course of action, it will be difficult for the critics to provide a comprehensive review of the proposed study.

Classification of Research Designs

The various designs have been classified by examining them from **three different perspectives**:

- Based on number of contacts with the study population
 - Cross sectional study design
 - Before and after study design
 - Longitudinal study design

- Based on the reference period of the study
 - Retrospective design
 - Prospective design
 - Retrospective-Prospective design
- Based on nature of investigation
 - Experimental design
 - Non-experimental design
 - Quasi-experimental design
- Based on objective of the study
 - Descriptive design: What is going on (descriptive research)?
 - Exploratory design
 - Explanatory design: Why is it going on (explanatory research)?

1. Based on the number of contacts:

The cross-sectional study design

- Cross-sectional studies, also known as one-shot or status studies are the most commonly used design in the social sciences.
- They are useful in obtaining an overall 'picture' as it stands at the time of the study. They are 'designed to study some phenomenon by taking a cross- section of **it at one time**
- A cross-sectional study is extremely simple in design. You decide what you want to find out, identify the study population, select sample (if you need to) and contact your respondents to find out the required information

The before-and-after study design

- The main advantage of the before-and-after design (also known as **the pre- test/post-test design**) is that it can measure change in a situation, phenomenon, issue, problem or attitude. It is the most appropriate design for measuring the impact or effectiveness of a program.
- A before-and-after design can be described as **two sets of cross-sectional data collection points** on the same population to find out the change in the phenomenon or variable(s) between two points in time. The change is measured by comparing the difference in the phenomenon or variable(s) before and after the intervention.

The longitudinal study design

- The before-and-after study design is appropriate for measuring the extent of change in a phenomenon, situation, problem, and attitude and so on, but is less helpful for studying the pattern of change.
- To determine the **pattern of change** in relation to time, a longitudinal design is used
- In longitudinal studies the study population is visited a number of times at regular intervals, usually over a long period, to collect the required information

2. Study Designs Based on the Reference Period

The reference period refers to the time-frame in which a study is exploring a phenomenon, situation, event or problem.

The retrospective study design

- Retrospective studies investigate a phenomenon, situation, problem or issue that has happened in the past. They are usually conducted either on the basis of the data available for that period or on the basis of respondents' recall of the situation.

The prospective study design

- The likely prevalence of a phenomenon, situation, problem, attitude or outcome in the future. Such studies attempt to establish the outcome of an event or what is likely to happen.

The retrospective-prospective study design

- Retrospective--prospective studies focus on past trends in a phenomenon and study it into the future. A study is classified under this category when you measure the impact of an intervention without having a control group.

1. Study Designs Based on the nature of the Investigation

Experimental Design

- In experimental design, researchers plan to measure the response variable depending on the explanatory variable. The response variable is an outcome measure for predicting or forecasting purposes of a study.
- It is also called dependent variable or predicted variable. Any variable that explains the response variable is called explanatory variable. It is also called independent variable or predictor variable
- A true experimental design is one in which study participants are randomly assigned to experimental and control groups. Although randomization is typically described using examples such as rolling dice, flipping a coin, or picking a number out of a hat, most studies now rely on

the use of random numbers tables to help them assign their research participants. The most important factor in the experimental design is randomization

- The *randomization check*, as its name suggests, is the process of examining the overall effectiveness of random assignment
- In performing randomization checks, researchers compare study
- Groups or conditions on a number of pretest variables. These typically include
- Demographic variables such as age, gender, level of education, and any other variables that are measured or available prior to the intervention.

Quasi-experimental Design

- Although the researcher plans to measure the response variable depending on the explanatory variable, there is a lack of randomization in the quasi-experimental design. It is a mixed design where random and non-random experiments are employed together.

2. Based on objective of the study

Descriptive research: its goal is description of the existing phenomena. It looks characteristics of individuals, groups, institutions in present state area of investigation to obtain precise information in various field of inquiry. It helps explain the study based on researchers knowledge.

Exploratory research: is used when little is known or no scientific knowledge is available about groups, activities, processes and situations.

Explanatory research: aims to understand phenomena by discovering and measuring causal relations among them and looks for causes and reasons.

Chapter Five

Sources And Methods Of Data Collections

Learning Objectives:

- identify sources of primary and secondary data
- distinguish advantage and disadvantage of primary and secondary sources
- discuss primary and secondary data collection techniques

What is data?

Data is a collection of raw facts and figures, such as numbers, words, measurements, observations or even just descriptions of things to be used in decision making.

Sources can be primary and secondary sources

1. Primary data

- Are collected afresh and for the first time, and thus happen to be original in character.
- Refers to information obtained firsthand by the researcher on the variables of interest for the specific purpose of the study. Four main primary sources of data; individuals, focus groups, panels, & unobtrusive methods:

1. Individuals: are people who provide information when interviewed, administered questionnaires, or observed.

2. Focus Groups: consist typically of 8 to 10 members with a moderator leading the discussions for about 2 hours on a particular topic, concept, or product. Provides only *qualitative* information and *not quantitative*. Members are generally chosen on the basis of their expertise in the topic on which information is sought. The moderator plays a vital role in steering the discussions in a manner that would draw out the information sought, and keeping the members on track.

Role of the Moderator

- The moderator introduces the topic, observes, and takes notes and/or tapes the discussions.
- The moderator never becomes an integral part of the discussions, but merely steers the group persuasively to obtain all the relevant information
- The moderator also ensures that all members participate in the discussion and that no member dominates the group.

3. Panels

- Panels are another source of primary information for research purposes.
- Whereas focus groups meet for a one - time group session, panels meet more than once.

➤ In cases where the effects of certain interventions or changes are to be studied over a period of time, panel studies are very useful. Panels can be:

1. **Static** - the same members serve on the panel over extended periods of time
2. **Dynamic** - the panel members change from time to time as various phases of the study are in progress

4. Unobtrusive Measures

Trace measures, or *unobtrusive measures* as they are also called, originate from a primary source that does not involve people. One example is the wear and tear of journals in a university library, which offers a good indication of their popularity, frequency of use, or both.

2. Secondary Data

- Data that are already available i.e., they refer to the data which have already been collected and analyzed by someone else.
- Any data that has been collected earlier for some other purpose are secondary data in the hands of an individual who is using them.
- Secondary data refer to information gathered from sources already existing.

Source of secondary data:

- ❖ *Company records or archives*
- ❖ *Government publications and Web sites*
- ❖ *Industry analyses offered by the media*
- ❖ *The Internet and so on.*

Data Collection Methods

Data can be collected in a variety of ways, most important data collection methods include:

1. **Interviews**- interview respondents to obtain information on the issues of interest.
- Interviews could be unstructured, structured and semi-structured, and conducted either face to face or by telephone or online.

Unstructured interviews - the interviewer does not enter the interview setting with a planned sequence of questions to be asked of the respondent. The objective of the unstructured interview is to bring some preliminary issues to the surface so that the researcher can determine what variables need further in-depth investigation.

Structured interviews- are those conducted when it is known at the outset what information is needed. The interviewer has a list of predetermined questions to be asked of the respondents either personally, through the telephone, or through the medium of a PC. The questions are likely

to focus on factors that had surfaced during the unstructured interviews and are considered relevant to the problem.

Semi-structured interviews- are a bit more relaxed than structured interviews. Researchers who use type are expected to cover every question in the protocol and have some wiggle room to explore participant responses by asking for clarification or additional information. Interviewers also have the freedom to be more friendly and sociable.

2. Questionnaires

Questionnaire is a pre-formulated written set of questions to which respondents record their answers. Questionnaires can be: - personally administered questionnaires, mailed questionnaires and electronically distributed.

Types of Questionnaires- are classified as Open-ended or closed ended type.

1-Closed ended questions:. include a set of questions to which respondents can reply in a limited number of ways.

2. Open ended questions: are unrestricted types of questionnaires that call for a free response in the respondents' own words. Often researchers use one of them exclusively or both of them in combination.

Principles of a Questionnaire

- Be clear and precise.
- Use natural and familiar language.
- Do not use words or phrases that show bias.
- Avoid double-barreled questions- Bad: "Do you like the taste and the color of this product?"
- State explicit alternatives.

3. Schedule- This method of data collection is very much like with questionnaire, with little difference, which lies in the fact that schedules are being filled in by the enumerators who are specially appointed for the purpose. These enumerators along with schedules go to respondents, put to them the questions from the proforma in the order the questions are listed and record the replies in the space meant for the same in the proforma.

4. Observation - It is possible to gather data without asking questions of respondents. People can be observed in their natural work environment or in the lab setting, and their activities and behaviors or other items of interest can be noted and recorded.

The researcher can play one of two roles while gathering field observational data.

Non-participant-Observer - The researcher collect the needed data in that capacity without becoming an integral part of the organizational system.

Participant-Observer- The researcher enters the organization and becomes a part of the work team. For instance, if a researcher wants to study group dynamics in work organizations, then she may join the organization as an employee and observe the dynamics in groups while being a part of the work organization and work groups.

Merits and Demerits of Data Collection Methods

Method	Advantages	Disadvantages
Interview (face to face)	It has the highest response rates.	very time consuming
	Personal contacts are involved	resource intensive
	Follow up questions can be asked.	Interviewer bias is greatest
Telephone	Cheaper than personal interview	Noise may interrupt
	Gather information rapidly	You may not have telephone number
Questionnaire	• lower costs and time	• Non-returns.
	• It is free from the bias of the interviewers	• Misinterpretation
	• standardization & privacy	• needs education & is inflexibility
Schedule	Suitable even for eliminates	Costly and time consuming
	No problem of response	Enumerator bias
Observation	Behavior is recorded as it occurs	Past behavior can't be observed
	Respondents willingness is not mandatory	Very slow
	Applicable for those who are not respond	Expensive

Advantages of Primary Data

- o Gives data in greater detail and More accurate
- o Includes definitions of terms and units
- o Investigators understand the meaning of units

Limitations of primary data

- o The use of primary data is relatively expensive than that of secondary data.
- o It relatively consumes more time and effort
- o It requires development of data collection instrument-which may be costly and time consuming.
- o May not appropriate when the research report is required urgently.

Advantages of Secondary Data

- More Economical and Saves time
- Prompt completion of report
- Improve understanding of the problem
- Used as a basis of comparison with primary data

Limitations of secondary data

- o may not fit our purpose
- o may not be similar to proposed study
- o difficulty to know accuracy
- o difficulty to know degree of care it had been collected
- o outdated

Chapter Six

Sample Design

Learning objectives: after studying this chapter you should be able to:

- define sampling
- understand the need for sampling
- characteristics of a good sampling design
- differentiate types of sample design

Important terms

1. Population- the entire group of people, events, or things of interest that the researcher wishes to investigate. if a banker is interested in investigating the saving habits of employees working in hotel industry of Debark town, then all hotel workers in Debark will form the population.

2. Element- is a single member of the population. If 500 hotel employees in a particular hotel happen to be the population of interest to a researcher, each hotel employees therein is an element.

3. Population Frame- is a listing of all the elements in the population from which the sample is drawn. The university registry containing a listing of all students in the university during a particular academic year could serve as the population frame for a study of a university students population.

4. Sample- is a subset of the population. It comprises some members selected from it. If hotel workers in Debark town are 1000 and if 200 members are drawn from a population of 1000 hotel workers, these 200 members form the sample for the study.

5. Subject- is a single member of the sample, just as an element is a single member of the population. If 200 members from the total population of 1,000 workers formed the sample for the study, then each hotel worker in the sample is a subject.

6. Sampling- is the *process* of selecting a sufficient number of elements from the population so that a study of the sample and an understanding of its properties would make it possible for us to generalize such properties to the population elements.

Reasons for Sampling- In research investigations, it would be practically impossible to collect data from every element because:-

- a) *Time, cost, and other human resources limitation.*
- b) Sometimes likely to produce more reliable results.

Characteristics of a Good Sampling Design

- Truly representative
- Having small sampling error
- Economically viable
- Systematic bias is controlled in a better way
- Results can be applied to the population in general with a reasonable level of confidence
- Optimum size (adequately large)
- Similar to population-should have all the characteristics that are present in the population and enable prediction of population characteristics based on sample

Types of Sampling Designs

I. Probability Sampling- When elements in the population have a known chance of being chosen as subjects in the sample, we resort to a probability sampling design.

The most common probability sampling designs are:-

1. Simple Random Sampling

In simple random sampling, every element in the population has a *known and equal* chance of being selected as a subject. **Let us say** there are 1,000 elements in the population, and we need a sample of 100. Suppose we were to drop pieces of paper in a hat, each bearing the name of one of the elements, and draw 100 of those from the hat with our eyes closed. We know that the first piece drawn will have a 1/1,000 chance of being drawn, the next one a 1/999 chance of being drawn, and so on.

2. Systematic Sampling

The systematic sampling design involves drawing every n^{th} element in the population starting with a randomly chosen element between 1 and n . *Sampling interval (n) is determined by dividing the population by the desired sample size.* **Example:** - If we want a sample of 35 households from a total population of 260 houses in a particular locality, then we could sample every seventh households starting from a random number from 1 to 7. Let us say that the random number is 7, then houses numbered 7, 14, 21, 28, and so on, would be sampled until the 35 houses are selected.

3. Stratified Random Sampling- Involves a process of stratification or segregation, followed by random selection of subjects from each stratum. The population is first divided into mutually exclusive groups that are relevant, appropriate, and meaningful in the context of the

study. Stratification ensures homogeneity within each stratum, but heterogeneity between strata. In other words, there will be more between group differences than within-group differences.

1. Proportionate stratified sampling- Members represented in the sample from each stratum will be proportionate to the total number of elements in the respective strata.

2. Disproportionate sampling- decisions are made either when some stratum or strata are too small or too large, or when there is more variability suspected or when the researcher believe that he/she will get more data within a particular stratum.

4. Cluster Sampling

Groups of elements that, ideally, would have heterogeneity among the members within each group are chosen for study in cluster sampling. When several groups with intra-group heterogeneity and intergroup homogeneity are found, then a random sampling of the clusters or groups can ideally be done and information gathered from each of the members in the randomly chosen clusters.

II. Non-probability Sampling

The elements in the population do not have any probabilities attached to their being chosen as sample subjects. The four very common types of non-probability sampling designs are:-

1. Convenience Sampling- refers to the collection of information from members of the population who are conveniently available to provide it. Example:- Students in a classroom can be considered to study how many students are missing classes in Debark University.

2. Judgment Sampling- involves the choice of subjects who are most advantageously placed or in the best position to provide the information required. The primary consideration in purposive sampling is your judgment as to who can provide the best information to achieve the objective of your study.

3. Quota Sampling- ensures that certain groups are adequately represented in the study through the assignment of a quota. Generally, the quota fixed for each subgroup is based on the total numbers of each group in the population. The actual selection of the items for the sample is left to the interviewer's discretion.

For example, consider a market research study where interviewers will stop people in the street to ask them a series of questions on consumer preferences. The interviewer might be asked to sample 200 people, of whom 100 should be male and 100 should be female - and, within each of

these groups, there should be 25 people in each of the age-groups: under-20, 20-39, 40-59 and over-60.

Respondents are selected not randomly but they are selected just because they are accessible to the interviewer.

4. Snowball Sampling- In this technique initial respondents are selected by probability or non-probability methods and then additional respondents are selected from information provided by the initial respondents. This method is used to locate members of rare population by referrals.

Sample Size Determination

In sampling analysis the most ticklish question is: What should be the size of the sample or how large or small should be 'n'? If the sample size ('n') is too small, it may not serve to achieve the objectives and if it is too large, we may incur huge cost and waste resources. As a general rule, one can say that the sample must be of an optimum size i.e., it should neither be excessively large nor too small.

Points to be considered in Determining Sample Size

- I. Number of classes proposed
- II. Nature of study
- III. Type of sampling
- IV. Precision
- V. Confidence level
- VI. Degree of variability

Methods of Determining the Sample Size

1. Arbitrary Approach: Uses a "rule of thumb" to determine the sample size. The researcher may determine, for instance, the sample size should be 5% of the population. Arbitrary sample size are simple and easy to apply, but they are not efficient and economical.

2. Conventional Approach: It follows some "convention", number believed somehow to be the right sample size. The convention might be an average of the sample sizes of similar studies.

3. Cost Analysis: It uses cost as a basis for sample size. In this case instead of the value of the information to be gained –from survey being a primary consideration in the sample size, the sample size is determined by cost factors that ignore the value.

4. Using Formulas to Calculate a Sample Size:

For infinite population (Kothari,1985:179)

$$\rightarrow n = \frac{z^2 * p * q}{e^2}$$

Where n= size of the sample, z= standard variety at a given confidence level, e is the desired level of precision, p is the estimated proportion of an attribute that is present in the population, and q is 1-p.

e.g Suppose we wish to evaluate a state-wide supporting program in which traders were encouraged to adopt a new practice. Assume that researcher wants to assess traders who effectively adopt the new practice but there is a large population and we do not know the variability population attributes. Furthermore, suppose we desire a 95% confidence level and $\pm 5\%$ precision. Find the resulting sample size.

$$n = \frac{(1.96)^2 (0.5) (0.5)}{(0.05)^2} = 385 \text{ traders}$$

For Finite Population

$$\rightarrow n = \frac{N}{1 + N(e)^2} \quad (\text{Yamane, 1967:886 as cited by Israel, 2013})$$

$$\rightarrow n = \frac{z^2 * p * q * N}{e^2 (N-1) + z^2 * p * q} \quad (\text{Kothari, 1985:179})$$

Where;

n= size of the sample,

N= the population size,

z= standard variety at a given confidence level,

e = the desired level of precision,

p= the estimated proportion of an attribute that is present in the population, and

q =1-p.

e.g Suppose we wish to evaluate a state-wide supporting program in which traders were encouraged to adopt a new practice. Suppose further that our evaluation of traders' adoption of the new practice only affected 2,000 traders and the intention of the study is to assess the effectiveness in the adoption of the new practice. Furthermore, suppose we desire a 95% confidence level and $\pm 5\%$ precision. Find the resulting sample size.

Sample size using the two formulas

$$n = \frac{2000}{1 + 2000(0.05)^2} = 333 \text{ traders}$$

$$n = \frac{(1.96)^2 (0.5)(0.5)(2000)}{(0.05)^2(2000-1) + (1.96)^2 (0.5)(0.5)} = 322 \text{ traders}$$

Chapter Seven

Data processing, analysis & interpretation

The data, after collection, has to be processed and analyzed in accordance with the outline laid down for the purpose at the time of developing the research plan.

Data Processing- Processing implies editing, coding, classification & tabulation of collected data so that they are amenable to analysis.

Data Analysis

The term analysis refers to the computation of certain measures along with searching for patterns of relationship that exist among data-groups. Data analysis is further transformation of the processed data to look for patterns and relations among data groups. Data Analysis is the process of systematically applying statistical and/or logical techniques to describe and illustrate, condense and recap, and evaluate the data. Analysis can be categorized as

❖ Descriptive Analysis and Inferential (Statistical) Analysis

1. Descriptive statistics

Descriptive statistics involve transformation of raw data into a form that would provide information to describe a set of factors in a situation. Descriptive analysis is largely the study of distribution of one variable. Analysis begins for most projects with some form of descriptive analysis to reduce the data into a summary format. Descriptive analysis refers to the transformation of raw data into a form that will make them easy to understand and interpret. Descriptive statistics are provided by frequencies, measures of central tendency, and dispersion.

Frequencies and Percentage: Table containing percentage and frequency distribution is easier to interpret. Percentages are useful for comparing the trend over time or among categories

Measure of central tendency: These measures are most useful when the purpose is to identify typical values of a variable or the most common characteristics of a group. Measure of central tendency is also known as statistical average. Mean, median and mode are most popular averages.

Mean (arithmetic mean) is the common measure of central tendency

Mode in such study like estimating the popular size of shoes it can be used

Median commonly used in estimating the average of qualitative phenomenon like estimating intelligence.

Measurement of dispersion: is a measurement how the value of an item scattered around the

true value of the average. Important measures of dispersion are:

Range: Measures the difference between the maximum and the minimum value of the observed variable

Mean deviation: is the average dispersion of an observation around the mean value. $\Sigma(X_i - \bar{X})/n$

Variance: It is mean square deviation. It measures the sample variability.

4. Inferential statistics - Is used to make inferences about the characteristics of a populations based on sample data. The goal is to go beyond the data at hand and make inferences about population parameters.

- the relationship between two variables (e.g., between advertisement and sales)correlation
- Differences in a variable among different subgroups (e.g., whether women or men buy more of a product) Regression
- How several independent variables might explain the variance in a dependent variable (e.g., how operational performances are influenced by human resources management (selection, training, compensation and promotion) Regression analysis (R^2)

Interpretation of Data

After collecting and analyzing the data, the researcher has to accomplish the task of drawing inferences followed by the report writing. Interpretation has to be done carefully so that misleading conclusion will not be drawn and the whole purpose of doing research will not be vitiated. Interpretation refers to the task of drawing inferences from the collected facts after an analytical and/or experimental study. The task of interpretation has two major aspects

- 1) The effort to establish continuity in research through linking the results of a given study with those of others.
- 2) The establishment of explanatory concept.

Technique of interpretation

The technique of interpretation often involves the following steps:-

- 1) Researcher must give reasonable explanation of the relation.
- 2) Extraneous information must be considered while interpreting the final result of research.
- 3) It is advisable to consult someone having insight into the study will result in correct interpretation and, thus, enhance the utility of research result.
- 4) Researcher must accomplish the task of interpretation only after considering all relevant factors affecting the problem to avoid false generalization.

Chapter Eight

Research Report Writing

Objective of this chapter: The purpose of this lesson is to give guidance in writing a research report which is simply the presentation of research findings to specific audience. After studying this lesson you will be able to:

- (i) Know the requirement of report writing.
- (ii) Understand the format of written report.
- (iii) Know the different referencing styles

Research Report Writing

Writhing report is the last step in a research study and requires a set of skills somewhat different from those called for in research of the earlier stages of research. This task should be accomplished by the researcher with at most care. He may also seek the assistance and guidance of experts for the purpose. The research task remains incomplete till the report has been presented and/or written.

Layout of the Research Report

The layout of the report means as to what the research report should contain. A comprehensive layout of the research report should comprise:

- (A) Preliminary pages
- (B) The main text
- (C) The end matters

(A) Preliminary Pages- Parts of the paper before chapter 1. In its preliminary pages the report should carry cover page, title page, acknowledgements, acronyms, table of contents, list of tables, abstract and so forth.

(B) Main Text- The main text provides the complete outline of the research report along with all details. Each main section of the report should begin on a new page. The main text of the report should have the following sections.

1. Chapter one- Introduction

- Background of the study
- statement of the problem
- objectives of the study
- Research Questions

- Significance of the study
- Delimitation of the study
- Definition of Terms (if any)
- Organization of the Study

2.Chapter two- Review of Related Literature

5. Research Methodology

- Description of the study area
- Population & Sampling Technique
- Types Data collected
- Methods of Data Collection
- Data Analysis Methods

6. Data Analysis and Interpretation

- 4.1. Characteristics of the Study Population
- 4.2. Analysis of the Findings of the Study

5. Summary, Conclusions and Recommendations

- 5.1. Conclusions
- 5.2. Recommendations

(C) End Matters

- References and Bibliography of sources consulted should be given.
- At the end of the report, Appendix should be enlisted in respect of all technical data